CS5701(Quantitative Data Analysis)

Lab 2: Independent Practice

The aim of this part of the lab exercise is to give you practical experience in working with confidence intervals in R Studio.

This independent practice is not assessed and does not need to be submitted - work through it at your own pace, and ask us (or the GTAs) for help at any point.

1 Confidence Intervals

1. Open R Studio and open a new project
2. Save the auto-mpg.csv file in the same folder as this new project
3. Open a new R Markdown file to use for this analysis
4. Read in the auto-mpg.csv data set into your new R markdown file
5. Explore the data (using summary statistics and plots) and check which columns from the continuous variables look normally distributed
6. Compute a 95% confidence interval for the mean for one of the variable that appears to be normally distributed
7. The model years range from the 70s and the 80s. What proportion of cars are from the 80s?
8. Compute a 90 % confidence interval for the proportion of cars from the 80s?
9. Change the confidence level and track what happens to the width of the confidence interval for 6. and 8.

Optional Extension A: Compute the confidence intervals for the mean using the "bootstrap method" (for details and sample code see p. 62 of *Statistics: an introduction using R*, M. Crawley)

About the data Dataset source: UCI Machine Learning Repository

- mpg: continuous
- cylinders: multi-valued discrete
- displacement: continuous
- horsepower: continuous
- weight: continuous
- acceleration: continuous
- model year: multi-valued discrete
- origin: multi-valued discrete
- car name: string (unique for each instance)

2 Distributions - Optional Extension B

1. Assume that among diabetics the fasting blood level of glucose is approximately normally distributed with a mean of 105 mg per 100 ml and a standard deviation of 9mg per 100ml.

   (a) Plot the density function for this distribution.

   (b) What proportion of diabetics have levels between 90 and 125 mg per 100ml? (This quantity is represented by the area under the normal curve between x=90 and x=125)

   (c) You should see from the plot that most of the area under the normal curve is contained between the two vertical lines. Therefore we should expect that the proportion of diabetics with levels between 90 and 125 mg per 100 ml to be high. Use the `pnorm` function to calculate this proportion. (you may need to use it twice)

   (d) What level cuts off the lower 10% of diabetics? (hint: use `qnorm`)

2. Going back to the `mpg` data set from Section 1: using the `acceleration` variable, overlay a fitted Normal density curve on top of its histogram, using the sample mean and standard deviation of `acceleration` as the parameters for `dnorm`. Does the fitted curve look like a good match for the shape of the histogram? Try the same for `weight` or `horsepower` - does the fit still look reasonable?

3 R code

You may find this code useful to complete this worksheet:

```
#to read in a csv file
x<-read.csv("filename.csv")

# to explore the contents of the data
summary(x)
# confidence interval for proportion
prop.test(x, n, p = NULL, conf.level = 0.95)

#confidence interval for mean
t.test(x,conf.level=0.9)
# for the binomial distribution
# the probability mass function for a given number of successes $v$, number of
    trials $n$ and probability of success $p$
dbinom(v,n,p)
#the cumulative distribution function for the same as above
pbinom(v,n,p)

#For the Normal distribution (the PDF) in order to find the y-values
    corresponding to a range of x values $(a,b)$
#with a mean of $m$ and a standard deviation of $sd$
dnorm(x, mean=m, sd=sd)
#the CDF for a given value of x
pnorm(x, mean, sd)
#the x value that corresponds to a given CDF
qnorm(p, mean, sd)
#for Z value the standard normal distribution x values:
qnorm(1- significance level)
```